Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/25/2020
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Clinically- and Spaceflight-Relevant Bacteria and Fungi and Drug Effectiveness under Simulated Lunar, Martian, and Microgravities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	Luis.Zea@Colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	407-242-2885
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Aerospace Engineering Sciences		
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PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	10/01/2018	**End Date:**	12/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
Contact Email:	Denise.E.Freeland@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 7/12/21) NOTE: End date changed to 3/31/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 9/30/2020 per NSSC information (Ed., 7/8/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stodieck, Louis Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	80NSSC18K1468		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbial (bacteria and fungi) behavior has been observed to change in microgravity: enhanced bacterial growth, improved cell aggregation, increased virulence, and reduced susceptibility to antibiotics have been reported on spaceflight experiments. However, these parameters have not been systematically assessed, and even less so at partial gravities: 1/6 g (lunar) and 1/3 g (Mars). This project aims to characterize changes on cell growth dynamics and cell aggregation as a function of these gravitational regimes (simulated micro-, 1/6, and 1/3 g). This will be complemented by gene expression analyses focused on the drug-resistance-associated genes (resistome) as well as those responsible for bacterial virulence. This systematic microbial study will include ten commensal, opportunistic, and pathogenic bacterial and fungal strains (of Escherichia coli, Pseudomonas aeruginosa, methicillin-resistant Staphylococcus aureus (MRSA), Candida albicans, Bacillus cereus, Staphylococcus haemolyticus, Salmonella typhimurium, and Shewanella oneidensis). This study has a second component that focuses on determining if there are changes on the concentrations of drugs needed to inhibit microbial growth at different gravitational regimes, like the ones expected to be experienced during deep space missions, the Moon, and Mars. For this study, four different antimicrobials (each representing distinct mechanisms of action) are planned to be used: gentamicin sulfate (aminoglycoside class), which has been used for three different space-based experiments; ciprofloxacin (fluoroquinolone class), the most commonly prescribed drug for urinary tract infections (UTI); Ceftolozane/Tazobactam (ß-lactam/ ß-lactam combination antibiotic used on drug-resistant infections); and Fluconazole (an azole-class antifungal). The use of spaceflight hardware, inclined clinostats to simulate partial- and microgravity, and human microbiome-associated bacterial and fungal models, will enable this project to address Risks and Gaps of NASA's Human Research Roadmap, as well as to expand our understanding of fundamental cellular and molecular mechanisms behind microbial responses to changes in gravity. Knowledge produced from this project may help us better prepare for human space exploration of other celestial bodies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying gene expression of bacteria and fungi exposed to antimicrobials under altered gravitational conditions may help unmask molecular genetic processes not observable under the 1 g environment on Earth. In turn, this may help scientists learn of how bacteria are able to develop resistance mechanisms and/or potentially find novel targets against drug-resistant bacteria.
Task Progress:	During the October 2019 - July 2020 timeframe, the following progress has been made: Literature produced: one book chapter (in review), one peer-reviewed journal manuscript (in review), one undergraduate honor's thesis, one conference paper, and one conference poster Escherichia coli AMG1 Growth in mAUMg hi under Various Simulated Gravities Background E.coli AMG01 Δ rpos Growth in mAUMg hi pi under Various Simulated Gravities Methicillin-resistant Staphylococcus aureus (MRSA) Growth in LB lennox under Various Simulated Gravities Salmonella typhimurium Growth in Medium E under Various Simulated Gravities Shewanella oneidensis Growth Curves in SBM at Simulated Reduced Gravities Pseudomonas aeruginosa PA14 Growth in mAUMg hi-Pi under Various Simulated Gravities Candida albicans at 1g in YBD Broth Minimum Inhibitory Concentration (MIC) for MRSA Grown Under 1g Conditions in LB Lennox Broth.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Books/Book Chapters	Zea L, Santa Maria SR, Ricco AJ. "CubeSats for microbiology and astrobiology research." in "CubeSat Handbook: From Mission Design to Operations." Ed. C. Cappelleti, S. Battistini, B. Malphrus. Cambridge, MA : Elsevier, 2021. p. 147-162. Available online 22nd September 2020. ISBN-13: 978-0128178843. https://doi.org/10.1016/B978-0-12-817884-3.00007-2 , Jan-2021
Dissertations and Theses	Forward T. "Preliminary Protocols for the Quantification of Shewanella oneidensis Iron Reduction Performance." Undergraduate Honors' Thesis, University of Colorado, Boulder, 2020. https://scholar.colorado.edu/concern/undergraduate_honors_theses/z029p569t , Apr-2020
Papers from Meeting Proceedings	Forward T, Allen L, Stodieck L. Klaus D, Zea L. "Growth Dynamics of Bacteria Under Simulated Lunar and Martian Gravities." 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. Paper IAC-19-A1.8.2x51510. https://iafastro.directory/iac/paper/id/51510/summary/ , Oct-2019